Cancer Mortality

Pre-Lab

Purpose: In this lab we will explore the relation between proximity to a nuclear materials storage site and cancer mortality rates.

Preview: We will learn how to model data, fit it to a straight line, interpret the model and measure how good our model is.

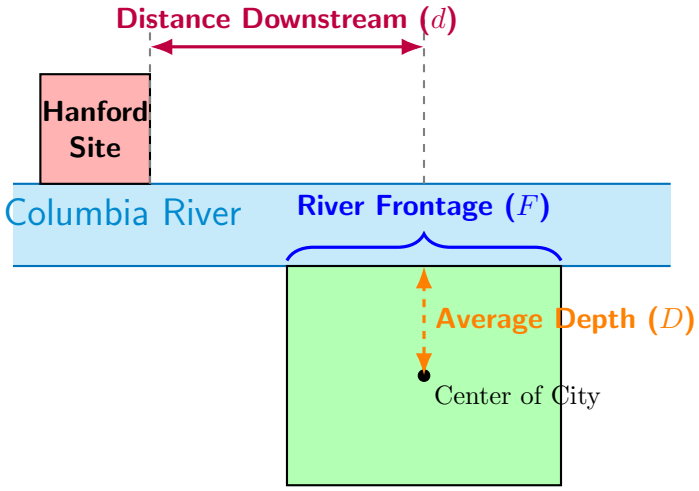
Background: Larsen and Marx (1986) state "Since World War II, plutonium for use in atomic weapons has been produced at an Atomic Energy Commission facility in Hanford, Washington. One of the major safety problems encountered there has been the storage of radioactive wastes. Over the years, significant quantities of these substances, including strontium 90 and cesium 137—have leaked from their open-pit storage areas into the nearby Columbia River, which flows along the Washington-Oregon border, and eventually empties into the Pacific Ocean."

To measure how exposure to the radiation affected cancer mortality rates during 1959-1964, researchers calculated an *index of exposure*. The index takes into consideration the county's distance down the river from the plant and the two measures of proximity of the county to the river.

Definitions

- **River Frontage:** The length of the county's (or city's) border along the Columbia River. That is, the length (in, say, miles) of the county's (or city's) waterfront along the river.
- **Distance from the Hanford site:** The distance downstream from the reactor to the center of the county (or city).
- **Average depth away from the river:** The distance from the center of the county (or city) to the nearest point on the Columbia River.

A diagram of all these parameters is shown below.



Mathematical Preliminaries

- y is *directly proportional* to x if there is a constant k such that $y = kx$.
- y is *inversely proportional* to x if there is a constant k such that $y = \frac{k}{x}$.
- y is *directly proportional* to x *and to* z if there is a constant k such that $y = kxz$.
- y is *directly proportional* to x *and inversely proportional to* z if there is a constant k such that $y = \frac{kx}{z}$.

The index of exposure is formulated on the assumption that county or city exposure is (Smith and Moore, 1996):

- directly proportional to the county's (or city's) *river frontage*;
- inversely proportional to the county's (or city's) *distance from the Hanford site*;
- inversely proportional to the square of the county's (or city's) *average depth away from the river*.

The following table gives the index of exposure and cancer mortalities (the average number of cancer deaths per 20,000 residents per year during the period 1959-1964).

County	Exposure	Mortality
Umatilla	2.49	147.1
Morrow	2.57	130.1
Gilliam	3.41	129.9
Sherman	1.25	113.5
Wasco	1.62	137.5
Hood River	3.83	162.3
Portland	11.64	207.5
Columbia	6.41	177.9
Clatsop	8.34	210.3

Questions

Take good notes on Questions 1 and 2! You'll need them to do Question 4.

1. Briefly explain why it makes sense that:
   - the index of exposure is directly proportional to river frontage (i.e why would a county with more river frontage be more exposed to radiation?);
   - the index of exposure is inversely proportional to distance from the Hanford site (i.e. why would a county further away from the Hanford site be less exposed to radiation?);
   - the index of exposure is inversely proportional to the square of average depth away from the river (i.e. why would a county whose center is further away from the river be less exposed to radiation?)[1].

[1] The reason we square this: it the most important factor in exposure, as it affects the entire county or city.

2. Let F be the river frontage, d be the distance from the Hanford site, and D be the average depth away from the river. By referring to the text above, write a formula for the index of exposure I in terms of F, d, and D. (You should use a constant of proportionality k in your formula.)

3. Open the spreadsheet for this lab here. Insert a scatter plot of Column C vs Column B. Make sure you customize your chart to start at $(0, 0)$[2]. *Don't add a linear trend line just yet.*

4. Open the pre-lab assignment for this lab in Gradescope and answer the multiple choice questions there.

Sources

1. Fadeley, R.C. (1965). Oregon malignancy pattern physiographically related to Hanford, Washington, Radioisotope Storage, Journal of Environmental Health 27, 883-897.

2. Larsen, R.J., and Marx, M.L. (1986). An Introduction to Mathematical Statistics and its Applications, 2nd Edition. Prentice-Hall, Englewood Cliffs, New Jersey. Case Study 1.2.4.

3. David A. Smith and L.C. Moore, Calculus: Modeling and Application, Houghton Mifflin Co., Single-Variable Chapters, 1996.

4. http://www.statsci.org/data/general/hanford.html

[2]Chart Editor → Customize → Horizontal axis, set 'Min' to 0. Then the same for the Vertical axis.

Main Lab

Part I: Modeling the Data

To start, have one member of your group share their spreadsheet with the rest of the group. Rename that sheet with all your group's names. Everyone in the group should have the sheet open on their laptops.

Also, create a Google doc (also shared with all group members) to record your answers to the questions below.

1. Why is it reasonable to model this data with a straight line?

2. What makes a good line to fit data? Write down a few criteria that characterize a good line.

3. - By picking two points so that a line between them might capture the trend of the data well, compute a slope and y-intercept for a line that might fit this data.
   - Insert your slope and y-intercept values in cells $B14$ and $B15$.
   - We are going to use these to compute y-values for your line in column D: y-intercept:
     - In cell $D2$, insert the formula `=B$14*B2+B$15`.
     - Fill down the formula to cell $D10$.
     - Briefly explain what this formula is doing. Specifically, what is the purpose of the $ symbols in the formula? And why doesn't the B2 term have a $ symbol in front of it? If you're not sure, try this:
       * Look at what's in cell D3 right now (and write it down).
       * Delete cells D2 through D10.
       * Re-insert the formula in D2, but this time without the $ symbols: `=B14*B2+B15`.
       * Fill down the formula to cell D10.
       * Look at what's in cell D3 right this time. Compare to what it was before. What is different? Why is this a problem? What did the dollar signs do?
       * Delete cells D2 through D10 again, and re-insert the formula in D2 with the $ symbols. Fill down to D10.
   - Now we will plot this line: Add the data in column D to your chart as a new series[3]. Make sure to customize the chart so that the new series is a line, not points[4]. You may want to change the color of the line so that it is easy to see.
   - Does your line fit as well as you thought (considering the criteria from the previous part)? If not, change the slope and y-intercept until it does.

4. Insert a linear trend line for the data on your chart. How does this new line compare to your answer in 3?

[3] Chart Editor → Setup. Change Data range to include column D, then add the series below.

[4] Chart Editor → Customize → Series. Choose only the series you just plotted. Set Point size to None. Turn on the trendline.

Part II: Interpreting the Model

For this part, use your equation from Part I, Question 3

5. What is the significance of the y-intercept in the equation? Is your answer reasonable, that is, does it make sense in the context of the situation?

6. What is the slope of your line? What is its significance in terms of the situation being modeled?

7. Estimate the mortality rate for a community with an index of exposure 7.0.

8. Estimate the indof exposure for a community with a mortality rate of 1000.

9. What numbers are reasonable values for the domain of this function? Explain.

Part III: A Measure of How Well a Line Fits the Data

In statistics, a common criterion for how well a line fits to a set of points is the sum of the squares of the vertical distances from the line to the data points.

10. What is an 'error' for a given line and a given data point? Why do we want to to square errors? Why do we want to sum them?

11. Is the following a correct formula for computing the sum of the squares of the vertical distances correct from a line $y = mx + b$ correct? If not, correct the formula.

$$\sum_{i=1}^{9}(y_i - (mx_i + b)),$$

where m is the slope of the line, b is the y-intercept, x_i is the index of exposure for the ith county or city, and y_i is the cancer mortality for the ith county or city.

12. In column E, insert the values predicted by your linear trend line (you may want to insert the slope and y-intercept values in $B18$ and $B19$). In columns F and G, compute the square differences for each of the two lines from Part I. At the bottom of each of those columns, sum the square differences to find the Sum of Square Errors for each of the lines. Compare your two answers. Do they make sense in comparison to one another? Explain.

Lab Report

For this report, you will use your understanding of the analysis carried out in lab to evaluate the claims made by a city along the river regarding their real estate development plan and its effects on cancer rates.

> "Our city, Columbia, has been experiencing a tragic rise in cancer mortality in recent times. Over the last five years, 4,900 people have died of cancer here. Having studied the Fadeley report, we strongly believe that a substantial number of these deaths were closely linked to the leakage of radioactive materials from the Hanford site. While data for cancer deaths prior to the construction of the nuclear facility are hard to come by, our data analysis leads us to believe that in the five years before the site was built, fewer than 2,900 died of cancer in our city. While some of the rise in cancer can be attributed to the growth in population in our town, such huge growth in deaths from this terrible disease must have an external cause.
>
> Over the past few months, we have come up with a plan to lower deaths from cancer in our beautiful city. We know from the Fadeley report that three factors contribute to our exposure to radiation: the length of our riverfront, our proximity to the Hanford site, and the distance of many of our houses to the river. While we cannot move our city, we can put a plan in place to affect the other factors.
>
> In the coming years, we will substantially increase tax credits for commercial, non-residential projects along our riverfront, focusing heavily on only half its current length. As residential projects are shut down, we will not issue new permits for them along the river. Instead, we will invest large amounts of money in construction of housing on the other side of town: far away from the polluted river. In this way, we will decrease our riverfront by half, and we project that the average distance of houses in our city from the river will increase by 10%. All this will take place over the next ten years. Despite projections that our town's population will grow by 15% over this time period, we believe this will lower the number of deaths from cancer substantially."

Throughout this report, you should use the linear regression model you found:

$$M = 9.23I + 115$$

where I is the index of exposure, and M is cancer mortality, with units defined above in the main part of the lab.

Recall from the introduction to the lab that the index of exposure is computed using

$$I = \frac{kF}{dD^2},$$

where F is river frontage, d distance from the reactor, and D average depth away from the river. k is a proportionality constant, whose value is not needed here.

You may use the model, the data table given on the first page, the data regarding cancer deaths in the plan, and the outcomes stated there. You may assume that the data given in the table on page 1 of the lab is current as of the time of the plan. For every question, you must explain your reasoning carefully in complete sentences and paragraphs, not just show computations.

Note: this report is designed to be a challenging assignment done together as a group over multiple days. You should seek support in Help Room from your lab TAs, as well as from your instructor in office hours or before/after class!

Questions

1. Suppose that the town succeeds in reducing the riverfront by a factor of 50%, and in increasing the average distance of the town's houses from the river by 10% over the next ten years.

   (a) Town planners claim that under this scenario, the new index of exposure will be approximately 2.65. Check their work using the formula for the index, showing all your work.

   (Hint: you know the current index from the data table, and you know the formula for the index. You need to compute the factor by which these changes will decrease the index.)

   (b) At the time of the report, the town has around 110,000 people. Assuming the town grows by 15% as stated, how many cancer deaths would be expected per year? Show all your work.

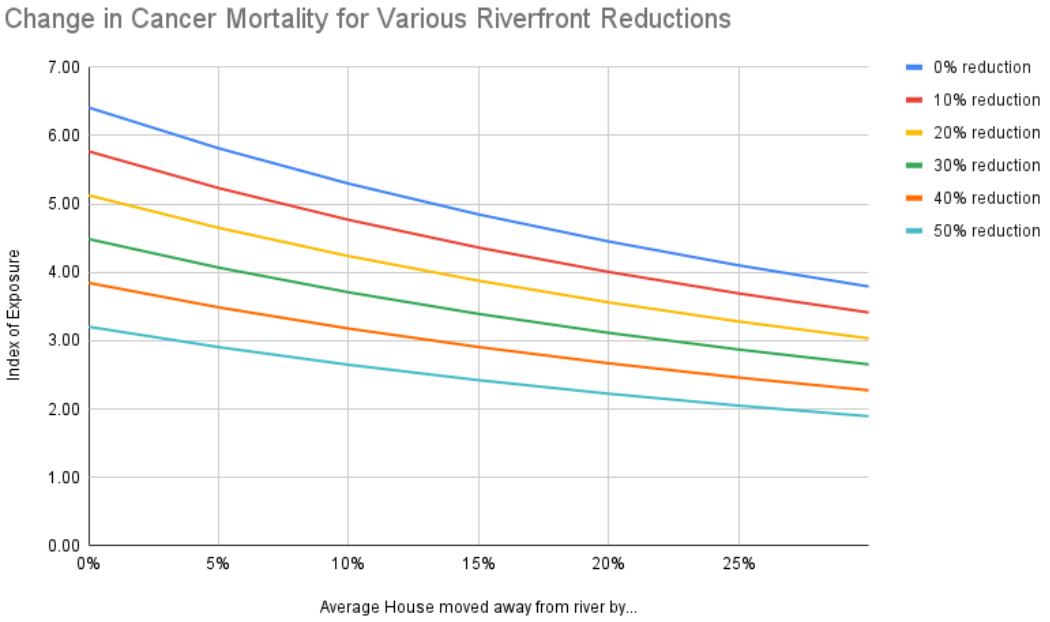
2. The mayor asked their team to project cancer death rates for a variety of reductions of the riverfront and moves away from the river. Below is the table of indices of exposure resulting from their computations:

Reduction in River Frontage

Avg Depth Increase	0%	10%	20%	30%	40%	50%
0%	6.41	5.77	5.13	4.49	3.85	3.21
5%	5.81	5.23	4.65	4.07	3.49	2.91
10%	5.30	4.77	4.24	3.71	3.18	2.65
15%	4.85	4.36	3.88	3.39	2.91	2.42
20%	4.45	4.01	3.56	3.12	2.67	2.23

   (a) Businesses along the riverfront complain to the City Council that a 50% decrease in river frontage would substantially hurt them. If river frontage were only to be cut by 30%, by how much would average depth away from the river have to increase to get the same effect on the index of exposure? Again, show all your work.

(b) Below is a chart showing this data:



i. How can you tell *from the chart* that the reduction in the index is not linear as the *average depth* increases?

ii. How can you tell the same thing just **from the formula for the index**?

(Hints: what do graphs of linear functions look like? What do formulas for linear functions look like?)

(c) The reduction in the index *is* linear as the *reduction in river frontage* increases. Explain why this is true from:

i. The formula for the index.

ii. The data in the table above. (Hint: look at the differences between the terms in each row!)

3. A town planner claims that since the index of exposure will be reduced from 6.41 to 2.65, a ratio of $\frac{6.41}{2.65} \approx 2.4$, the rate of cancer mortality per 20,000 population per year will also decrease by this ratio.

(a) Do you agree? Explain your answer carefully, using mathematical reasoning throughout. (Hint: compute the expected Cancer death rate/20k people/yr in each case.)

(b) Define *excess cancer mortality* to be:

$$\text{Excess Mortality} = \text{Rate of Cancer Mortality} - \text{Background Rate of Cancer Mortality}$$

measured in cancer deaths per 20,000 people per year. Compute excess mortality for an index of 6.41 and for an index of 2.65. Compute the ratio between your two answers. (Hint: recall that the background cancer rate is the number of cancer deaths/20k ppl/yr when the index of exposure is zero.)

(c) The mayor, who is running for reelection, claims the following in a debate:

> "My real estate plan will reduce our town's exposure from the very high index of 6.41, which has caused so many deaths, to a low 2.42, a nearly 60% reduction! This will reduce cancer deaths in our dear city by 60% as well."

As the mayor's opponent, what would be your response? Write a short paragraph, using your analysis in parts (a) and (b). Be sure to tailor your answer to the median voter, who knows not much math, and has never heard the term 'excess cancer deaths'!

(Hint: why won't a reduction of 60% in exposure reduce cancer deaths by the same amount? Why isn't the number of cancer deaths with zero exposure zero?)